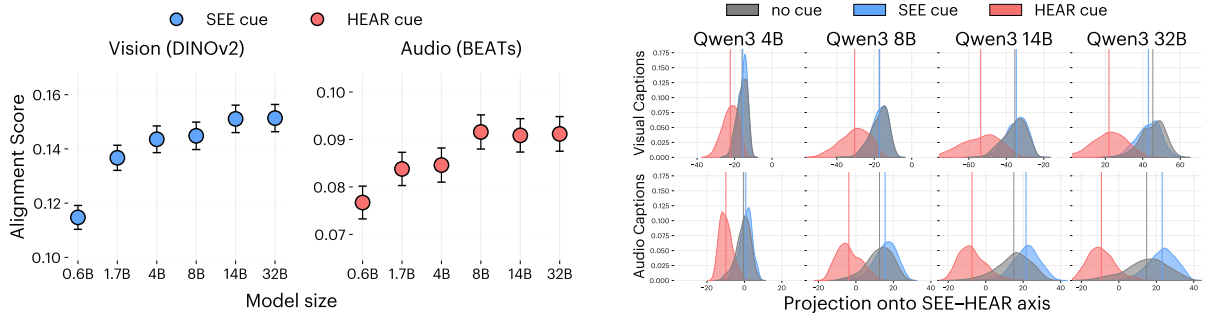

(a) Larger models are more aligned to sensory encoders under corresponding cues.

(b) Embedding projections onto visual–auditory axis show clearer modality separation in larger models.

Figure 8: Stronger sensory alignments and modality separations emerge in larger models.

Controlling for hallucinations. However, sensory language itself is not sufficient. Mutual-kNN captures relational similarity: it evaluates whether a caption and an image (or audio) induce similar neighborhoods over a dataset. To show that observed alignment gains do not arise from "hallucinations," where generic modality-specific words are added rather than attributes that accurately describe the given sample, we edit captions with additional visual words. That is, the mere presence of such visual descriptors could not increase alignment to a vision encoder. We sampled 10 random visual attributes from the 45,092 object properties parsed from Visual Genome [Krishna et al., 2017], and constructed variants of each caption from WiT in the form `{caption}` → `{caption + 10 random visual words}` and `{caption}` → `{10 random visual words}`. In Figure 7b, we find that alignment decreases when captions are appended with random visual words, and drops further when captions are replaced entirely by them. This indicates that the observed gains do not simply arise from hallucinations of modality-specific vocabulary, but instead reflect that mutual-kNN captures relational structure tied to scene-appropriate sensory detail. That is, sensory cues steer LLMs toward a correct modality-specific generation that brings caption–caption relations in the LLM closer to those of vision or audio models.

3.4 Generation Length Improves Sensory Alignment

In Figure 6, we find that alignment to vision and audio encoders increases as the LLM generates more tokens, suggesting that longer outputs give the model more opportunity to elaborate modality-specific content. At shorter lengths (32 and 64 tokens), the LLM often produces little beyond restating the prompt (e.g., 'Okay, the user wants me to imagine…'). Longer generations allow the model to reason about the scene.

Interestingly, even mismatched cues (e.g., SEE prompts evaluated on audio alignment) can outperform the no cue baseline. For instance, at 256 tokens, Qwen3-32B achieves better alignment to both modalities under either cue than under a neutral prompt. We interpret this as an effect of shared cross-modal structure: many sounds (e.g., snoring, barking) have associated visual features, so visually descriptive generations can still increase alignment with auditory encoders. However, we note that alignment can decline as you continue to increase output tokens due to semantic drift from the prompt (Appendix A).

3.5 Sensory Alignment Scales with Larger Models

We find that larger models have higher alignment with vision and audio encoders under appropriate sensory cues (Figure 8a; 128-token generations). Moreover, cue-specific representations become more separable with scale (Figure 8b). To quantify this, we project embeddings $\mathbf{x}_i \in \mathbb{R}^d$ onto a sensory axis defined by the mean difference between prompt conditions. Let $\mu_{\text{SEE}} = \frac{1}{N}\sum_i x_i^{\text{SEE}}$, $\mu_{\text{HEAR}} = \frac{1}{N}\sum_i x_i^{\text{HEAR}}$, be the mean embeddings under each cue. We define $\mathbf{v} = \frac{\mu_{\text{SEE}} - \mu_{\text{HEAR}}}{\|\mu_{\text{SEE}} - \mu_{\text{HEAR}}\|}$, and compute projections $s_i = \mathbf{x}_i^\top \mathbf{v}$, giving a scalar position along the visual–auditory axis. We estimate the distribution of s_i using kernel density estimation.

This result reflects more separate modality-specific representations with increasing scale (see Figure 29 for the same evaluation on DCI). In smaller models, no cue generated embeddings consistently resemble those generated by SEE, even when the caption describes a sound, suggesting that language models tend to default to a visual framing without an explicit cue. As model size increases, however, no cue embeddings shift closer to those generated by HEAR.

Figure 9: Instead of answering from an `(image, Q)` pair as in standard VQA, the model receives a `(caption, Q)` pair, where the caption is a text projection of the image.

Table 1: Visual prompting applied to the MME benchmark projected to text.

	artwork	celebrity	code_reasoning	color	commonsense	count	existence	landmark	numerical	position	posters	scene	text_translation	Overall
No cue	60.50	50.29	95.00	81.67	81.43	70.00	85.00	53.50	80.00	51.67	70.07	70.75	**97.50**	64.78
SEE	**61.50**	**51.76**	**97.50**	81.67	**82.86**	**75.00**	**90.00**	**54.75**	80.00	**55.00**	**78.91**	**72.25**	92.50	**67.14**
n	400	340	40	60	140	60	60	400	40	60	294	400	40	2334

3.6 Visual Question Answering in Text Space

We ask whether sensory prompting meaningfully creates better sensory representations in terms of downstream sensory task performance. To answer this, we test whether sensory cues allow language models to perform more accurate visual reasoning in the text modality, we adopt the "VQA without V" setting from Chan et al. [2025]; Chai et al. [2024]. Instead of providing an (image, question) pair as in standard VQA, we provide a (caption, question) pair, where captions serve as projections of images into text space. The model is then tasked with answering yes/no questions based solely on these captions. We use the MME benchmark [Fu et al., 2023a] and first caption all images with Qwen2.5-VL-3B-Instruct. Because OCR questions require recognizing text directly from the image rather than reasoning over the caption, we exclude the OCR category from our evaluation. We then evaluate Qwen3-14B as the question-answering model under two prompt conditions: a neutral instruction and a visual framing, which explicitly asks the model to *imagine seeing* the caption before answering. The full prompt can be found in Appendix F.

The visual framing yields a small but significant overall gain (64.78 $\rightarrow$ 67.14, $n = 2334$; see Table 1). MME categories falling under "Cognition (Reasoning Tasks)", including *numerical_calculation* and *text_translation*, do not experience improvement. These findings support the view that language models can act as text-space vision models, and that simple sensory cues improve performance specifically where reasoning about the imagery helps disambiguate the caption.

4 Related Work

Alignment between LLMs and models trained on modalities grounded in sensory data has been observed several times in past work, even though LLMs only experience the world through text. Abdou et al. [2021] demonstrate that the geometry of color word embeddings in LLMs aligns with human perception of these colors. Patel & Pavlick [2022] show that text-only LLMs can generalize structured concepts from the physical world, such as spatial directions, allowing them to reason about navigation with terms like "left" and "right" despite never having direct perceptual experience. These results are consistent with the observation that models trained on different modalities converge in their representation as the models scale [Huh et al., 2024].

Notably, some LLM representations do not capture sensory structure. Xu et al. [2025] show that while text-only LLMs can represent abstract properties of words, they fall short on sensory and motor features: when asked to rate words, their responses matched human ratings best in non-sensorimotor dimensions. However, Pavlick [2023] argue that the lack of direct grounding does not mean LLMs are unable to represent meaning. That is to say, weak sensory representations do not rule out the possibility of better ones. Our results demonstrate one such case: sensory cues can steer generative representations toward the intended modality, even though direct

embeddings do not show the same effect.

Gu et al. [2023] show that models can learn to solve visual tasks using only language supervision, demonstrating that captions can act as proxies for images. This connects directly to our VQA experiment, where we test whether sensory cues improve text-only models' ability to reason about captions as if they were images. Ashutosh et al. [2025] further show that through iterative feedback from a vision/audio model, a text-only LLM can perform multimodal captioning and generation.

5 Discussion

Perception involves both the reception of stimuli through sensors and their interpretation in context. We have shown that cueing a text-only language model to imagine a specific modality shifts its internal representation toward that of an explicit sensory encoder, which has a representation reflective of the true sensory structure because it is trained on that modality. When prompted to "see" (or "hear") the model behaves as if its input were grounded in perceptual evidence—producing modality-appropriate responses shaped by the generation of accurate visual (or auditory) imagery. This is quantified by comparing the kernel induced over captions by the LLM to the kernel induced by a sensory encoder over paired data; higher alignment under sensory prompting means that caption–caption relationships are more like those in the vision (or audio) model.

Our work extends prior observations of passive cross-modal convergence by showing that alignment can also be actively steered at inference time. This supports a view of language models as implicitly multimodal agents: their representations encode a distribution over possible latent causes, including sensory ones, for the text they process. Importantly, generative representations allow us to induce a kernel over a set of text inputs according to a prior specified in the prompt. In our case the prior is sensory, but in principle it could be other characteristics—for example, spatial layout or sentiment. This means prompting allows an interpretable way to steer which relationships the kernel encodes, rather than leaving them implicit to the model.

Sensory prompting also has a practical implication: it gives us an interpretable way to extract multimodal embeddings from models trained only on text. These embeddings could support tasks such as cross-modal retrieval, evaluation, or distilling knowledge into smaller sensory encoders. More broadly, it suggests that the line between unimodal and multimodal models is less rigid than often assumed—under the right prompts, an LLM can act like an image or audio encoder that operates in the text modality. Sensory prompting also fits alongside chain-of-thought and retrieval cues as part of a growing toolkit for inference-time control, showing that what a model represents is not fixed at training but can be elicited through context.

6 Limitations

While we have determined that text-based sensory cues can increase representational alignment to vision and audio encoders, we have not fully explored the degree to which this alignment can be improved. In particular, we focus primarily on lightweight cues such as 'see' and 'hear', but do not explore broader variation in instruction phrasing that could better elicit perceptual framing. We evaluate sensory prompting with null prompts and with other verbs in Appendix A. Furthermore, we note that LLM alignment to audio encoders is lower than alignment to vision encoders. One explanation is that audio encoders like BEATs learn low-level acoustic patterns (e.g., frequency, rhythm, timbre) that map less directly to language, whereas vision encoders such as DINOv2 capture object- and scene-level features that align more naturally with words. Supporting this, BEATs variants fine-tuned with AudioSet labels achieve much higher alignment (see Appendix C). Finally, sensory prompts may encourage the model to hallucinate specific perceptual details in the generation that are not actually supported by the input. While this is acceptable—and even desirable—in generative contexts, it may be problematic in settings requiring factual visual or auditory precision.